

LABORATORY AUTOMATION SOLUTION



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This research proposes a laboratory automation solution to optimise the workflow in clinical laboratories, leveraging cutting-edge technologies and techniques in human-robot collaboration and machine learning.

PROJECT STAKEHOLDERS

Industry Partner: Prof Peter O'Loughlin – SA Pathology
Supervisor: Dr Ivan Lee

PROJECT OUTLINE

This project is a laboratory design for SA Pathology and RAH that migrate the traditional system into automated system to **improve efficacy and precision while ensuring the cost-effectiveness of the workflow.**

In laboratory environment setting, automation solution integrate robot in assisting human with low value - high repetitive tasks that **improved overall performance by 66%** while **reducing human hazardous activities by 62%** [1].

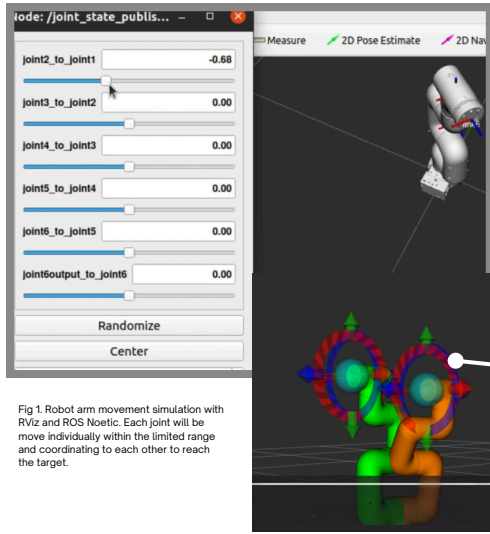


Fig 1. Robot arm movement simulation with RViz and ROS Noetic. Each joint will be move individually within the limited range and coordinating to each other to reach the target.

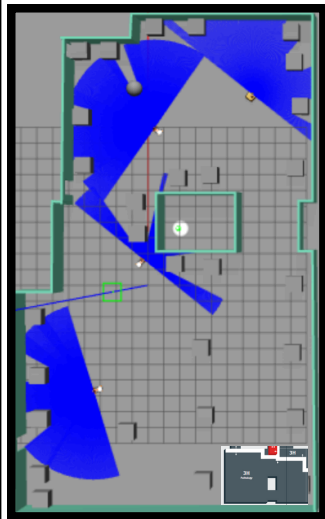


Fig 2. Mobile robot simulation in a laboratory environment with mapping and localization using SLAM (based on a real map of RAH pathology lab). Each AMR (Automatic mobile robot) will be equipped with a sensor and a dynamic map of the area to navigate around. These are representative of KUKA Mobile Robot that will be suggested in this research.

APPROACH

The system will be divided into three sections, each is responsible for a specific type of procedure in clinical processing:

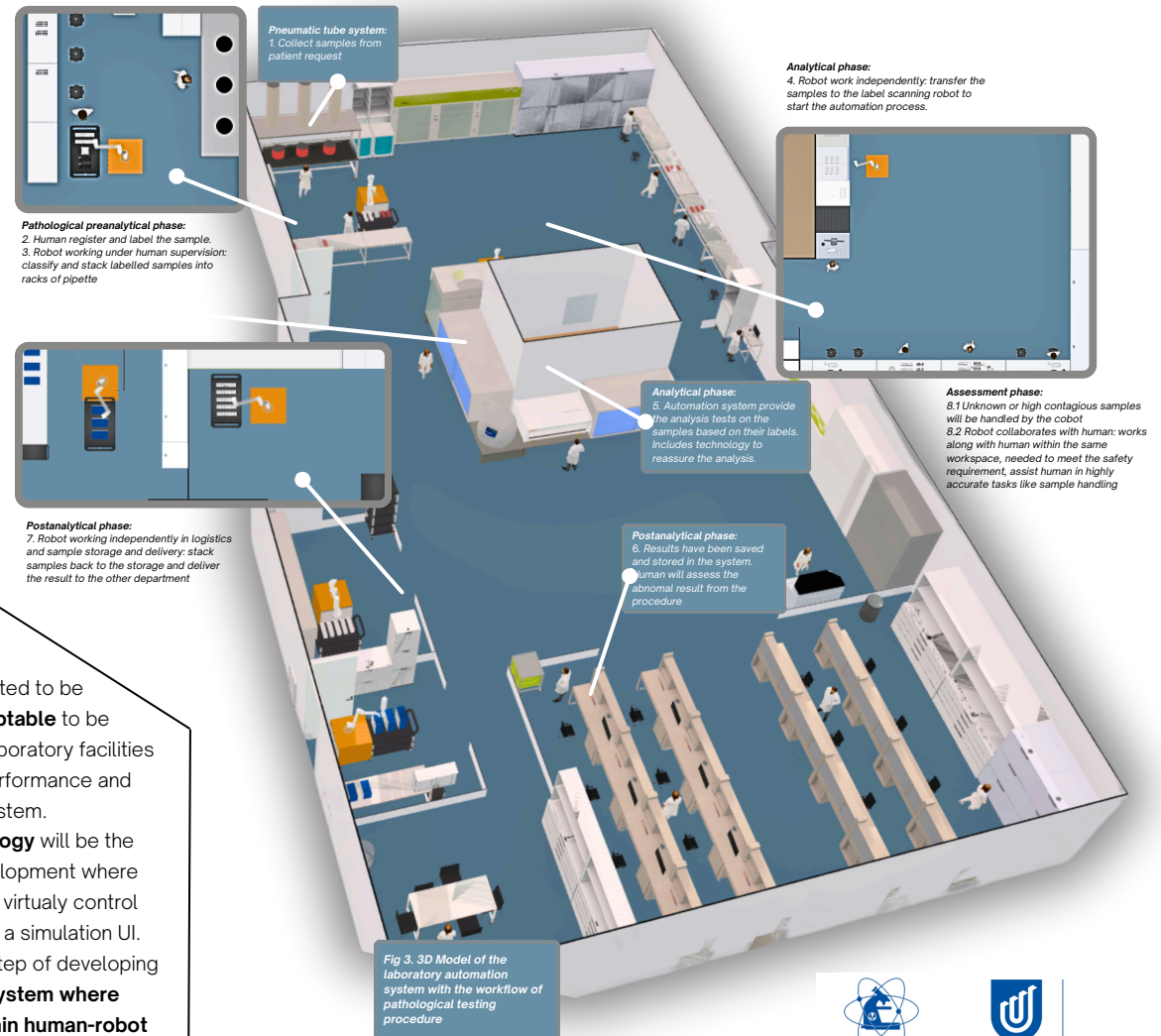
- **Robot works independently:** performs low value - high repetitive tasks
- **Robot collaborates with human:** perform tasks that dangerous for human or require high accuracy.
- **Robot works under human supervision:** perform tasks that require human monitoring to ensure proper operation and handle exceptions

The system will be simulated using **Gazebo and ROS** to imitate the robot behaviour in laboratory setting.

FUTURE DIRECTION

The solution is expected to be **extendable and adaptable** to be applied in different laboratory facilities while ensuring the performance and productivity of the system.

Digital Twin Technology will be the next step of the development where human will be able to virtually control the physical robot via a simulation UI. This will be the first step of developing a **total automation system where trust calibration within human-robot interaction would not be an issue.**



[1] ANDREA MARIA, Z & FEDERICA, F 2022, 'A COLLABORATIVE ROBOTIC SOLUTION TO PARTLY AUTOMATE SARS-COV-2 SEROLOGICAL TESTS IN SMALL FACILITIES', SLAS TECHNOLOGY, VOL. 27, NO. 1, PP. 100-106.